

versions, called alleles. There may be two to several hundred alleles of a gene, although each person can only have two. This variation in alleles accounts for the differences between individuals, such as eye color and ear shape. A dominant allele may “overpower” a recessive one (see [Dominant and recessive genes](#)).

Why genetic disorders occur

Sometimes a gene is faulty. Genetic disorders arise either when an abnormal gene is inherited or when a gene changes, or mutates. Genetic disorders may follow a [dominant or recessive](#) pattern of inheritance. They can also be passed on via the X chromosome. Such sex-linked disorders are usually recessive, which means that a woman can carry the faulty gene without being affected, because she has another healthy X chromosome to compensate. If a boy receives an affected X chromosome, he will be affected; a girl will be a healthy carrier like her mother. An affected male could pass on the affected gene only to his daughters.